Sabrim Mahmud

Researcher in multilingual NLP, AI systems developer

Dhaka, Bangladesh mahmudsabrim@gmail.com +880 1754 971710 github.com/Sabrimmahmud
linkedin.com/in/sabrim-mahmud-048b88313

Objective

I want to build a career in academia as a Computer Science educator and researcher: teaching well, mentoring students, and doing research in artificial intelligence, machine learning and natural language processing that matters outside the lab. My research and software engineering experience help me connect theory with working systems.

Education

Bachelor of Science in Computer Science and Engineering Jan 2022 – Jan 2026
Daffodil International University, Dhaka, Bangladesh
CGPA 3.84 / 4.00, 148 of 148 credits completed
Undergraduate thesis, industrial R&D project: *UniSentix: A Multilingual and Multimodal Sentiment Analysis Framework with Explainable Insights*
Selected coursework: Natural Language Processing, Artificial Intelligence, Data Mining and Machine Learning, Introduction to Data Science, Statistics and Probability, Algorithms, Data Structures, Database Management Systems, Software Engineering, System Analysis and Design, Research and Innovation in CSE, Web Engineering, Compiler Design, Computer Networks.

Higher Secondary Certificate (HSC) 2020
Rajshahi Govt. City College, Rajshahi, Bangladesh
Science group, GPA 5.00 / 5.00

Secondary School Certificate (SSC) 2017
Akkelpur F. U. Pilot High School
Science group, GPA 5.00 / 5.00

Peer-reviewed publications

[1] PoliticaNet: A Transformer-Driven Framework for Fine-Grained Classification of Bangla Political Hate Speech in Social Media

Sabrim Mahmud, Samaun Rezvi Payel, Jawad Hossain. *Asia Pacific Advanced Network (APANConf 2025)*, Springer, 2026, Communications in Computer and Information Science, vol. 2837, pp. 93–111. DOI: 10.1007/978-3-032-18319-4_6. The publisher notes equal contributions.

[2] Identifying Bangla Political Hate Speech using Transformer-based Approach

Sabrim Mahmud, Samaun Rezvi Payel, Jawad Hossain, Nuzhat Tabassum Progga, Asma Mariam. *2025 2nd International Conference on Next-Generation Computing, IoT and Machine Learning (NCIM)*, IEEE, 2025, pp. 1–6. DOI: 10.1109/NCIM65934.2025.11159848.

Presentations

Author and paper presenter, NCIM 2025 27–28 June 2025
Presented “Identifying Bangla Political Hate Speech using Transformer-based Approach”, Gazipur, Bangladesh.

Research experience

Research Assistant & AI Developer 2024 – 2025
Daffodil International University, AR/VR Lab, Dhaka, Bangladesh

- **AI research.** Worked on hate speech detection, cyberbullying classification and multilingual NLP research.
- **Medical machine learning.** Developed MRI classification pipelines with EfficientNet, YOLO segmentation, and continual learning with EWC and LwF.
- **Full-stack systems.** Built database-driven academic systems and real-time applications.
- **Experimentation and evaluation.** Contributed to AI model experimentation, evaluation and research-based software development.

Industry experience

AI Systems Developer Aug 2025 – 2026
Enovosoft, Bangladesh
Built a multilingual sentiment platform for Bangla, English and Banglish with transformer-based NLP: the model integration, a unified API, an analytics dashboard, forecasting, and quantized local inference.

Mentorship and service

Program Mentor, Daffodil ModelX, Daffodil AI Club, Daffodil International University 2026
Mentored participants in the Daffodil ModelX programme, supporting their learning and skill development.

Research & Development (R&D) Team Member, Faculty-supervised R&D team, Daffodil International University
Took part in a faculty-supervised research and development team that later became affiliated with Daffodil AI Club.

Awards and competitions

Paper presentation, NCIM 2025 27–28 June 2025
Paper presenter. Department of CSE, Dhaka University of Engineering & Technology, Gazipur

Data Visionary: National Data Analytics Competition 2025 12 April 2025
Participant. Department of CSE, Daffodil International University

Crack Dataset 2024 2024
Fifth place. Department of CSE and NLP & ML Research Lab, Daffodil International University

Research interests

Natural language processing, Multilingual and low-resource NLP, Transformer models, Explainable artificial intelligence, Responsible and trustworthy AI, Hate speech detection, Deep learning and applied machine learning, AI for accessibility and healthcare, Multimodal AI.

Skills

AI and ML	PyTorch, Transformers and fine-tuning, Natural language processing, ONNX, Quantization (bnb-4bit, GGUF), LLM integration (Qwen), Time-series forecasting, EfficientNet and YOLO, Continual learning (EWC, LwF)
Programming	Python, TypeScript, JavaScript, C++, Java, SQL